

NAVNEET MANOJ

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Dual-degree student in Mechanical Engineering and Physics at BITS Pilani with hands-on experience in Python-based data analysis, multivariate statistical modelling, and sensor/instrument data pipelines. Comfortable working across the full analysis cycle — from raw measurement data to insight — with a strong interest in applying data science to human performance and physiological systems.

EDUCATION

BITS Pilani, Hyderabad Campus

2022 – 2027 (Expected)

B.E. Mechanical Engineering + M.Sc. Physics (Dual Degree)

CGPA: 7.48

Kuriakose Elias English Medium School, Mannanam

2022

Class XII (ICSE) — 93.7%

Harisri Vidya Nidhi, Punnamm, Thrissur

2020

Class X (ICSE) — 94.0%

TECHNICAL SKILLS

Data Analysis & ML: Python (NumPy, Matplotlib, Pandas), R, multivariate ANOVA, Grey Relational Analysis, Genetic Algorithm optimisation

Statistical Methods: Full-factorial DOE (L27), regression, multi-objective optimisation, POC algorithm validation

Simulation & CAD: SRIM/TRIM, MATLAB, AutoCAD, SolidWorks, Abaqus, X-Foil

Instrumentation: UV-Vis spectroscopy, photoluminescence analysis, thermocouple-based heat measurement

Programming: Python, C, C++, Pytorch(Deep Learning)

RESEARCH PROJECTS

Drilling Optimisation of PMMA Bone Cement — Multivariate Statistical Analysis

Aug 2025 – Present

- Designed an L27 full-factorial experiment varying drill diameter, feed rate, and spindle speed; applied ANOVA to decompose variance and identify dominant factors driving heat generation — drill diameter accounted for the largest temperature contribution.
- Executed Grey Relational Analysis for multi-objective optimisation across correlated response variables (temperature, machining time), achieving a best-case of 56.9 °C in 20 s — safely below tissue necrosis thresholds.
- Delivered a parametric guideline by integrating experimental data, statistical modelling, and process-level optimisation — a proof-of-concept for algorithm-driven decision support in physical systems.

Airfoil Shape Optimisation using Genetic Algorithm

Oct 2025 – Jan 2026

- Implemented a genetic-algorithm optimisation framework in Python — encoding airfoil geometry parameters (thickness, camber, taper, aspect ratio, twist) as a chromosome and evolving populations via selection, crossover, and mutation operators.
- Integrated Prandtl Lifting Line Theory and a vortex panel method as the fitness evaluation function, combining induced and viscous drag to assess each candidate at a fixed target lift coefficient.
- Demonstrated convergence to optimised solutions, establishing a lightweight and generalisable supervised-search tool for parametric design — directly applicable to any domain requiring structured parameter-space exploration.

Solar Cell Radiation Tolerance Simulation for Space Applications

Dec 2025 – Present

- Built a Python post-processing pipeline to parse multi-output simulation data (VACANCY, COLLISION, EXYZ files) from SRIM/TRIM irradiation runs on CsSnI₃ perovskites, generating vacancy density profiles and defect-depth visualisations.
- Applied statistical analysis to identify a damage peak and extract vacancies per ion; structured the analysis as a reusable proof-of-concept framework for radiation tolerance assessment.

Continuous Synthesis and Characterization of Quantum Dots in Microchannels

Dec 2024 – Present

- Designed and executed a controlled microfluidic synthesis pipeline for carbon quantum dots, systematically varying temperature, residence time, and precursor ratios — applying multivariate parameter analysis to optimise fluorescence output.
- Processed UV-Vis and photoluminescence spectroscopy data to extract optical performance metrics; demonstrated hydrothermal-comparable results at half reaction time (~100 s), validating the experimental protocol through quantitative comparison.
- Developed Python scripts to automate data parsing, statistical comparison, and visualisation of optical measurements across experimental conditions.

PATENTS & PUBLICATIONS

Patent Submitted: *Continuous Synthesis of Carbon Quantum Dots in Microchannels* (Application Submitted, BITS Pilani)

- CompFlu 2024** (Dec 2025) — "*Continuous Synthesis and Characterization of Carbon Quantum Dots in Microchannels*" [Poster]
- CACM** (2026) — "*Continuous Synthesis of Carbon Quantum Dots using Microfluidic Channels*" [Oral]

- ICFTE (2026) — "Airfoil Shape Optimisation using Genetic Algorithm" [Oral]

INTERNSHIP EXPERIENCE

Mechanical Engineering Intern — IFB Industries Limited, Goa

May 2024 – Jul 2024

- Designed and prototyped an Active Noise Cancellation (ANC) circuit using electret microphones, 2N2222A transistor pre-amplification, and LM386 audio amplification to attenuate washing-machine operational noise in real time — gaining hands-on experience in signal capture, conditioning, and real-time measurement systems.

RELEVANT COURSEWORK

Computational Physics · Statistical Mechanics · Quantum Mechanics · Principles of Aerodynamics · Heat Transfer · Fluid Mechanics · Mechanics of Solids · Material Science · Solid State Physics · Nuclear Physics · Electromagnetic Theory · Electromagnetic and Optics Lab

EXTRACURRICULARS & LANGUAGES

- Goalkeeper, BITS Pilani Hyderabad Football Team; active in tennis, basketball, cricket, swimming, and athletics.

Languages: English, Hindi, Tamil, Malayalam